1 Chern Classes

1.1 Grassmannian Example

Let $V = k^4$ and $W \subseteq V$ be such that $\dim W = 2$. Then $\mathbb{P}^1_k = \mathbb{P}(W) \subseteq \mathbb{P}(V) = \mathbb{P}^3_k$. We can identify W with a 2×4 matrix

$$M_W = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

such that $W = \text{col}M_W$. Note that $w \in W$ if and only if the surjective evaluation $w^* : V \to k$ corresponding to w factors through $M_W^\intercal : V \to W$; that is, the following diagram can be filled in so that it commutes:

$$\begin{array}{ccc} V & \xrightarrow{M_W^\intercal} & W \\ & \searrow^{w^*} & \downarrow \\ & & k \end{array}$$

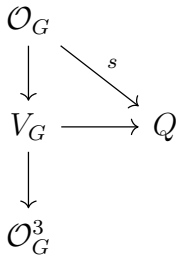
For example, consider $w = \begin{pmatrix} 2 \\ 5 \\ 0 \\ 0 \end{pmatrix}$, then $w^* = \begin{pmatrix} 2 & 5 & 0 & 0 \end{pmatrix} : V \to k$, and $w^* M_W^\intercal(v) = w^* v$ for all $v \in V$.

Now let $G = \mathfrak{Gr}(2,4)$ be the grassmannian of projective lines in $\mathbb{P}^3$, and let $V_G = V \times G$ be the trivial bundle of rank four on G. There is a tautological bundle of rank 2 on G whose fiber over the point $\ell \in G$ corresponding to W is W; call this bundle Q. Note that $Q \ncong W \times G$: two lines in $\mathbb{P}^3$ may intersect. There is a surjective map $V_G \to Q$ which– over the point $[M_W : V \to W] \in G$– maps the fibers of the bundles by M_W. That is, a fiber of $V_G|_{M_W}$ is $V \times \{M_W\}$, and a fiber of $Q|_{M_W}$ is $W \times \{M_W\}$, and the map of fibers $V_G|_{M_W} \to Q|_{M_W}$ is simply the map $M_W : V \times \{M_W\} \to W \times \{M_W\}$.

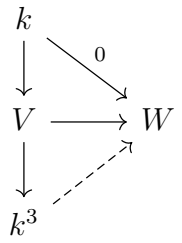
Let's find the Chern classes of Q. To fine $c_2(Q)$, we'll take a single section of Q and look at its zero locus. Start by picking a 2-plane $\Pi \cong \mathbb{P}^2 \subseteq \mathbb{P}^3$. This corresponds to a surjection $M_\Pi : V \to k^3$, so for each line $M_W : V \to W$, we get a diagram

$$\begin{array}{ccc} k & \searrow & \\ \downarrow & & \\ V & \longrightarrow & W \\ \downarrow & & \\ k^3 & & \end{array}$$

Globalizing, we have a similar diagram for bundles,



This map s is the section Q whose zero locus we will take to represent $c_2(Q)$. Over a given $M_W : V \to W$ in G, the section is zero if and only if there is a factorization



That is,

$$\operatorname{img}(k \to V) \subseteq \ker(M_W^{\mathsf{T}}) = \left\langle \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\rangle,$$

so

$$\operatorname{img}(k \to V) = \left\langle \begin{pmatrix} 0 \\ 0 \\ a \\ b \end{pmatrix} \right\rangle.$$

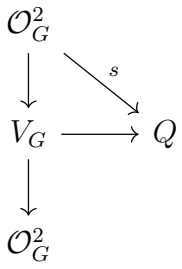
Here is where the $k^3 \to W$ map comes from: the short sequence

$$k \to V \to k^3$$

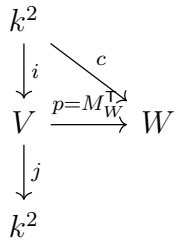
is supposed to by exact, so $V \to k^3$ is projection onto $\operatorname{img}(k)^\perp$, which contains W. This means the line $\mathbb{P}^1 = \mathbb{P}(W)$ is contained in the 2-plane $\Pi = \mathbb{P}(k^3)$. Therefore $\text{Zeros}(s) = \{\ell \in \mathbb{P}^3 : \ell \subseteq \Pi\} = \Sigma_2(\Pi)$. To compute $c_1(Q)$, we'll find two sections of Q and find the degeneracy locus. Start by picking a line $\ell_0 \subseteq \mathbb{P}^3$; this corresponds to the short exact sequence

$$0 \to k^2 \to V \to k^2 \to 0.$$

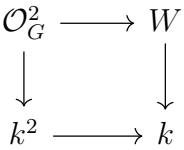
Globalizing, we get a map of bundles on G:



The diagonal arrow (composition of $\mathcal{O}_G^2 \to V_G \to Q$) gives our two sections of Q. Where does this map drop rank? At $\ell = \mathbb{P}(W) \in G$, the map of fibers is



where j corresponds to ℓ_0. If $\operatorname{rank}(c) < 2$, then $\ker(M_W^\mathsf{T}) + \ker(V \to k^2)$ is at most 3-dimensional in V. If it is 2-dimensional, then in fact $\ell = \ell_0$; otherwise it is 3-dimensional and we can fill in the diagram



showing ℓ meets ℓ_0. Therefore $c_1(Q) = \Sigma_1(\ell_0)$.

Also the first Chern class of the tangent bundle of the Grassmannian $Gr(k, n)$ is $c_1 = n\sigma_1$; this is consistent with the canonical divisor being $-n\sigma_1$.

1.2 Formulae

We have the Whitney sum formula: if

$$0 \to A \to B \to C \to 0$$

is a short exact sequence of vector bundles, then the Chern polynomial of B is

$$c(B) = c(A)c(C).$$

Now suppose $X \subseteq Y \subseteq Z$ is a sequence of bundles. Then there is a short exact sequence

$$0 \to X \to Y \to Y/X \to 0,$$

so by Whitney,

$$c(Y) = c(X)c(Y/X).$$

Additionally, there is a short exact sequence

$$0 \to Y \to Z \to Z/Y$$

and so

$$c(Z) = c(Y)c(Z/Y) = c(X)c(Y/X)c(Z/Y).$$

By induction, whenever $E_1 \subseteq E_2 \subseteq \cdots \subseteq E_n$ is a chain of bundles,

$$c(E_n) = c(E_1)c(E_2/E_1)\cdots c(E_n/E_{n-1}).$$

Zach also says (page 10) that $c_m(T_{\mathbb{P}^n}) = \binom{n+1}{m}H^m$ (note that $c_1(T_X) = -K_X$). We can write

$$T_{\mathbb{P}^n} = L_1 \oplus L_2 \oplus \cdots \oplus L_n \oplus L_{n+1}$$

where $c(L_i) = 1 + H$ for all i. Then

$$c(T_{\mathbb{P}^n}) = (1+H)^{n+1} = \sum_{m=0}^{n+1} \binom{n+1}{m} H^m = \sum_{m=0}^{n} \binom{n+1}{m} H^m,$$

since $H^{n+1} = 0$.

We can try analyzing this from first principles for $\mathbb{P}^2$. We wish to show that $c_1(T_{\mathbb{P}^2}) = 3L$ and $c_2(T_{\mathbb{P}^2}) = 3L^2$. The tangent bundle is trivial at a cubic curve, since the cubic curve is an abelian variety and thus has a trivial tangent bundle itself.

Consider a twisted cubic $X \subseteq \mathbb{P}^3$ on a quadric surface Q. WLOG let $X \sim 2\alpha+\beta \in A(Q)$. Then we have

$$0 \to T_X \to T_{\mathbb{P}^3}|_X \to N_{X/\mathbb{P}^3} \to 0,$$

where $c(T_{\mathbb{P}^3}|_X) = [1 + 4H + 6H^2 + 4H^3] \cap X = 1 + 12P$. We also know that $c(T_X) = 1 + c_1(T_X) = 1 - ([-2(\alpha+\beta) + (2\alpha+\beta)].(2\alpha+\beta)) = 1 + 2P$, by the adjunction formula. And so

$$c(N_X) = \left.\frac{1+12P}{1+2P}\right|_X = 1 + 10P,$$

which is consistent with the literature (Jesse Leo Kass' notes).

Also we can do

$$0 \to T_X \to T_Q|_X \to N_{X/Q} \to 0$$

and

$$0 \to T_Q \to T_{\mathbb{P}^3}|_Q \to N_{Q/\mathbb{P}^3} \to 0$$

where $c(N_{Q/\mathbb{P}^3}) = 1+2H_Q = 1+2\alpha+2\beta$ and $c(T_{\mathbb{P}^3}|_Q) = 1+4H_Q+6H_Q^2 = 1+4\alpha+4\beta+12\alpha\beta$ and so $c(T_Q) = c(T_{\mathbb{P}^3}|_Q)/c(N_{Q/\mathbb{P}^3}) = 1+2\alpha+2\beta+4\alpha\beta$.

Then next $c(T_Q|_X) = 1+2\alpha_X+2\beta_X+0 = 1+2P+4P = 1+6P$. By the adjunction formula, $c(T_X) = 1+2P$, and so $c(N_{X/Q}) = 1+4P$, differing from $c(N_{X/\mathbb{P}^3}) = 1+10P$.

2 Blowups

2.1 Line in $\mathbb{P}^3$

Let L be a line in $\mathbb{P}^3$ and take X to be the blowup of $\mathbb{P}^3$ along L. Then

$$A(X) = \mathbb{Z}[h,e]/(h^4, (h-e)^2, (he)(h-e)-h^3).$$

Then

$$h^2 = 2he - e^2,$$

or

$$e^2 = 2he - h^2.$$

Furthermore,

$$h^3 = he^2 - e^3,$$

or

$$e^3 = he^2 - h^3 = h(2he-h^2) - h^3 = 2h^2e - 2h^3 = -2h^3.$$

Actually, it turns out $h^4 = 0$ is redundant. This is consistent with Eisenbud and Harris' description that

$$A(X) = \mathbb{Z}[\alpha,\zeta]/(\alpha^2, \zeta^3 - \alpha\zeta^2)$$

under the identifications

$$e = \zeta - \alpha$$

and

$$h - e = \alpha,$$

that is,

$$\zeta = h.$$

2.2 Twisted Cubic

Let $\zeta \in A^1(E)$ denote the first Chern class of $\mathcal{O}_{\mathbb{P}\mathcal{N}}(1)$, which is $10P$, as from Section 2.2.

Consider the diagram

$$\begin{array}{ccc} \mathrm{Bl}_C(\mathbb{P}^3) & \xrightarrow{\pi} & \mathbb{P}^3 \\ {\scriptstyle j}\uparrow & & \uparrow{\scriptstyle i} \\ E_C & \xrightarrow{\pi|_{E_C}} & C \end{array}.$$

We can construct the Chow ring $A(\mathrm{Bl}_C(\mathbb{P}^3))$ as being generated by $\pi^*(A(\mathbb{P}^3))$ and $j_*(A(E_C))$ under the following rules:

- $\pi^*\alpha \cdot \pi^*\beta = \pi^*(\alpha\beta)$ for $\alpha, \beta \in A(\mathbb{P}^3)$
- $\pi^*\alpha \cdot j_*\gamma = j_*(\gamma \cdot \pi|^*_{E_C} i^*\alpha)$ for $\alpha \in A(\mathbb{P}^3)$ and $\gamma \in A(E_C)$
- $j_*\gamma \cdot j_*\delta = -j_*(\gamma\delta\zeta)$ for $\gamma, \delta \in A(E_C)$.

In this case

$$\zeta = \pi|^*_{E_C}(10P),$$

as from Section 2.2. Now we need to understand $A(E_C)$. We have

$$A(E_C) \cong \mathbb{Z}[w, z]/(w^2, z^2 + 10wz),$$

wherein w represents the class of a fibre: $w = \pi|^*_{E_C}(P)$, and z represents the class of a section, and wz represents the class of a point in E_C. That is,

$$z^2 = -\zeta z.$$

So now we have enough information to find $A(\mathrm{Bl}_C(\mathbb{P}^3))$. Note we have the short exact sequence

$$0 \to A_\ell(C) \to A_\ell(E_C) \oplus A_\ell(\mathbb{P}^3) \to A_\ell(\mathrm{Bl}_C(\mathbb{P}^3)) \to 0,$$

for every integer ℓ. We also have

$$0 \to \mathcal{O}(-1) \to N_{C/\mathbb{P}^3} \to N_{E_C/\mathrm{Bl}_C(\mathbb{P}^3)} \to 0,$$

so $c(N_{E_C/\mathrm{Bl}_C(\mathbb{P}^3)}) = (1 + 10w)/(1 - z) = (1 + 10w + z)$. So now we have the complete Chow ring as follows:

0. A^0 is freely generated by 1
1. A^1 is freely generated by H and E
2. A^2 is generated by W, Z, and H^2, where $10W + Z = 3H^2$
3. A^3 is freely generated by H^3

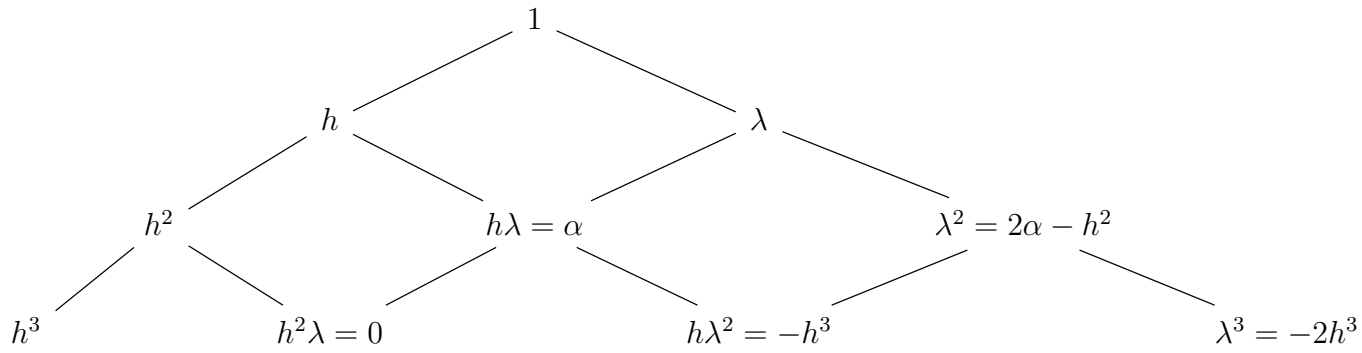
2.3 Lightning Round

There is a typo in Eisenbud and Harris' *3264 and all that intersection theory*, where they say in Proposition 13.13 that $\deg(e.F_D) = \deg D$ but it should in fact be $-\deg D$.

Example 1. Blow up $\mathbb{P}^3$ at a line. Then

$$\begin{aligned} A^0 &= \langle 1 \rangle \\ A^1 &= \langle h, \lambda \rangle \\ A^2 &= \langle h^2, h\lambda = \alpha, \lambda^2 \rangle \\ A^3 &= \langle h^3 \rangle \end{aligned}$$

where $\lambda^2 = -j_*\zeta = 2\alpha - h^2$. Furthermore, $\lambda\alpha = -h^3$. We can arrange the monomials as follows.



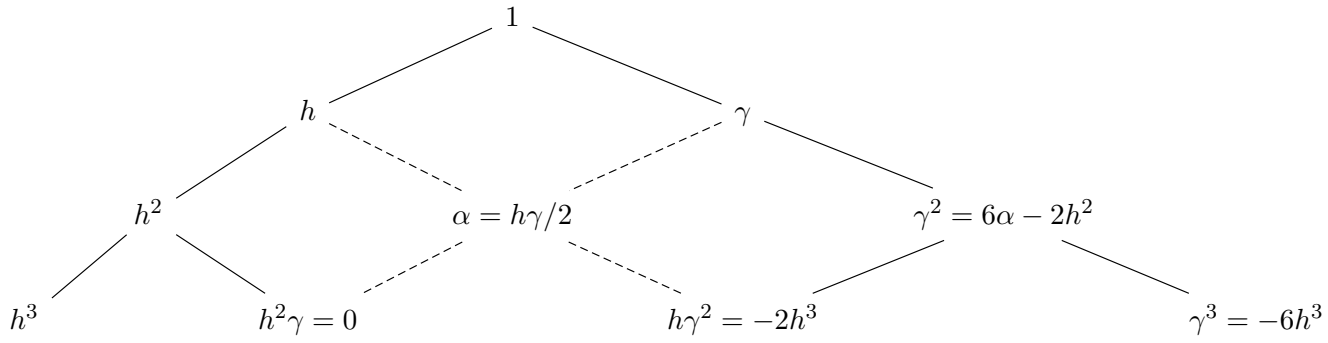
We can also define $\beta = (h-\lambda)\lambda = \alpha - (2\alpha - h^2) = h^2 - \alpha$. We can also confirm $(h-\lambda)^2 = h^2 - 2h\lambda + \lambda^2 = h^2 - 2\alpha + 2\alpha - h^2 = 0$, and $(h-\lambda)h\lambda = h\alpha - \lambda\alpha = 0 - -h^3 = h^3$.

Also $h^2 - 2\alpha = -\lambda^2$, while also $h^2 - 2\alpha = h^2 - 2h\lambda = h(h - 2\lambda)$.

Example 2. Now let's blow up $\mathbb{P}^3$ at a conic. We get

$$\begin{aligned} A^0 &= \langle 1 \rangle \\ A^1 &= \langle h, \gamma \rangle \\ A^2 &= \langle h^2, \alpha, \gamma^2 \rangle \\ A^3 &= \langle h^3 \rangle \end{aligned}$$

where $\gamma\alpha = -h^3$, $h\alpha = 0$, $h^2\gamma = 0$, $h\gamma^2 = -2h^3$, and $\gamma^3 = -6h^3$. We also have $\gamma^2 = -j_*(\zeta) = 6\alpha - 2h^2$. Note that $6\alpha = c_1(\mathcal{N})$ and $2h^2$ is the class of a conic in $\mathbb{P}^3$. So $j_*(\zeta)$ is given by the first Chern class of $\mathbb{P}(\mathcal{N})$ 6α as well as the degree of the curve being blown up $2h^2$. We can arrange the monomials as follows.



We can also define $\beta_1 = (h-\gamma)\gamma = h\gamma - \gamma^2 = 2\alpha - (6\alpha - 2h^2) = 2h^2 - 4\alpha$ and $\beta_2 = (2h-\gamma)\gamma = 2h\gamma - \gamma^2 = 4\alpha - (6\alpha - 2h^2) = 2h^2 - 2\alpha$. Also $(h-\gamma)^2 = h^2 - 2h\gamma + \gamma^2 = h^2 - 4\alpha + 6\alpha - 2h^2 = 2\alpha - h^2 = -2\beta_1$, and $(2h-\gamma)^2 = 4h^2 - 4h\gamma + \gamma^2 = 4h^2 - 8\alpha + 6\alpha - 2h^2 = 2h^2 - 2\alpha = \beta_2$.

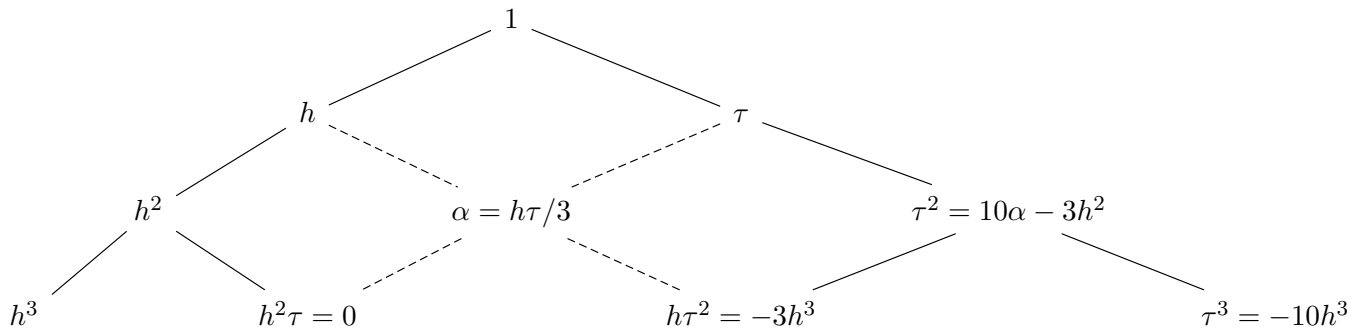
Also $(h-\gamma)(2h-\gamma) = 2h^2 - 3h\gamma + \gamma^2 = 2h^2 - 6\alpha + 6\alpha - 2h^2 = 0$.

Also $2h^2 - 5\alpha = 2h^2 - 10h\gamma = 2h(h-5\gamma)$, while also $2h^2 - 5\alpha = -\gamma^2 + \alpha = -\gamma^2 + h\gamma/2 = (h/2-\gamma)\gamma$.

Example 3. Now let's go back to a twisted cubic in $\mathbb{P}^3$. We get

$$\begin{aligned} A^0 &= \langle 1 \rangle \\ A^1 &= \langle h, \tau \rangle \\ A^2 &= \langle h^2, \alpha, \tau^2 \rangle \\ A^3 &= \langle h^3 \rangle \end{aligned}$$

where $\tau\alpha = -h^3$, $h\alpha = 0$, $h^2\tau = 0$, $h\tau^2 = -3h^3$, and $\tau^3 = -10h^3$. Again, $\tau^2 = -j_*\zeta = c_1(N)\alpha - 3h^3 = 10\alpha - 3h^2$. Also, $h\tau = 3\alpha$. We can arrange the monomials as follows.



Note that $(2h-\tau)^2 = 4h^2 - 4h\tau + \tau^2 = 4h^2 - 12\alpha + 10\alpha - 3h^2 = h^2 - 2\alpha$. That is because two quadrics containing a twisted cubic will meet at the cubic and a line. WLOG let the cubic have bidegree-(1,2) on the quadric; then the line will have bidegree-(1,0), so the line will meet the twisted cubic at two points, hence the -2α term.

We also have $(2h-\tau)^3 = (h^2 - 2\alpha)(2h-\tau) = 2h^3 - h^2\tau - 4\alpha h + 2\alpha\tau = 2h^3 - h^3 = h^3$, one point. So I suppose there is one point/normal vector pair on the twisted cubic that all three quadric surfaces have in common?

We can also define $\beta = (2h-\tau)\tau = 2h\tau - \tau^2 = 6\alpha - 10\alpha + 3h^2 = 3h^2 - 4\alpha$.

Also $3h^2 - 6\alpha = -\tau^2 + 4\alpha = -\tau^2 + 12h\tau = (12h-\tau)\tau$, also $3h^2 - 6\alpha = 3h^2 - 18h\tau = 3h(h-6\tau)$.

2.4 A Point and a Line in $\mathbb{P}^3$

Let X be the blowup of $\mathbb{P}^3$ at a point. Then

$$A(X) \cong \mathbb{Z}[h,\pi]/(h^4, (h-\pi)^3, h\pi).$$

Note that this implies $(h-\pi)^2\pi = h^3$ which makes sense. Then let Y be the blowup of X at a line L. Then we have three situations to analyze: the first, if $L \cap \pi = \varnothing$; the second, if $L \cap \pi$ is a point; and finally, if $L \subseteq \pi$. The first situation is the easiest to handle: we simply have

$$A(Y) \cong \mathbb{Z}[h,\pi,\lambda]/(h^4, (h-\pi)^3, h\pi, (h-\lambda)^2, h\lambda(h-\lambda)-h^3, \pi\lambda).$$

The second situation is also quite manageable:

$$A(Y) \cong \mathbb{Z}[h,\pi,\lambda]/(h^4, (h-\pi)^3, h\pi, (h-\pi-\lambda)^2, h\lambda(h-\pi-\lambda)-h^3, (h-\pi-\lambda)(h-\pi)\pi-h^3).$$

We needn't concern ourselves with the final case.

2.5 Tetrahedral Blowup

Now we can describe the blowup Z of $\mathbb{P}^3$ at four points and six lines in a tetrahedron. We can identify four π-type exceptional loci $\pi_1, \dots, \pi_4$, and six λ-type exceptional loci $\lambda_{12}, \lambda_{13}, \lambda_{14}, \lambda_{23}, \lambda_{24}, \lambda_{34}$. Then

$$\mathbb{Z}[h, \pi_i, \lambda_{jk}] \twoheadrightarrow A(Z),$$

under the relations h^4, $(h-\pi_i)^3$, $h\pi_i$, $(h-\pi_j-\pi_k-\lambda_{jk})^2$, $h\lambda_{jk}(h-\pi_j-\pi_k-\lambda_{jk})-h^3$, and

$$\begin{cases} (h-\pi_j-\pi_k-\lambda_{jk})(h-\pi_i)\pi_i - h^3 & \text{if } i \in \{j,k\}, \\ \lambda_{jk}\pi_i & \text{else.} \end{cases}$$

Don't forget $\lambda_{ij}\lambda_{k\ell} = 0$ and $\pi_i\pi_j = 0$ too! The following involution is the Cremona transformation.

$$\begin{pmatrix} 3 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -2 & 0 & -1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -2 & -1 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -2 & -1 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -2 & -1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 1 \\ -1 & 0 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 1 & 0 \\ -1 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ -1 & -1 & 0 & 0 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ -1 & -1 & 0 & -1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & -1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

This is corroborated by the observation that the Cremona involution of a general plane has a double point at each of the base points. This matrix fixes the vector

$$\begin{pmatrix} -4 & 2 & 2 & 2 & 2 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}^{\mathsf{T}},$$

which is the canonical divisor.

3 Weyl Theory of these Blowups

Here is the part where this document becomes notes on the Weyl Theory papers I've collected :)

Let $T_{p,q,r}$ be a graph of $(p-1)+(q-1)+(r-1)+1 = p+q+r-2$ vertices containing one central vertex with three branches of length $p-1$, $q-1$, and $r-1$. It defines a lattice (a free abelian group equipped with an integral quadratic form) $Q_{p,q,r}$ of rank $p+q+r-2$ with a basis $e_1, \ldots, e_{p+q+r-2}$ corresponding to the vertices and

$$(e_i, e_j) = \begin{cases} -2 & i = j, \\ 1 & e_i, e_j \text{ are joined by an edge,} \\ 0 & \text{otherwise.} \end{cases}$$

The orthogonal transformations $s_i : x \mapsto x + (x, e_i)e_i$ generate a subgroup W of the orthogonal group $O(Q_{p,q,r})$.

Let V be a real vector space and V^* its dual space of linear functions. A (symmetric) root base in V is a finite set $B = \{\alpha_1, \ldots, \alpha_r\}$ of linearly independent vectors in V together with an injective map $\vee : B \to V^*$ such that the images $\check{\alpha}_i$ are linearly independent, $\check{\alpha}_i(\alpha_i) = -2$, $\check{\alpha}_i(\alpha_j) = \check{\alpha}_j(\alpha_i)$ for $i \neq j$ are nonnegative integers. The Dynkin diagram of a root base is a nonorientable graph whose vertices are the vectors α_i and the vertices α_i and α_j are joined by $\check{\alpha}_i(\alpha_j)$ edges. The lattice $Q(B)$ is called the root lattice on B. The adjacency matrix is called the **Cartan matrix** of B. The s_i maps are the **simple reflections**, which generate the **Weyl group** W. An s-translate $s \in W$ is called a **root**, the set of which is denoted R. The set

$$C = \{x \in V : \check{\alpha}_i(x) > 0, 1 \leq i \leq r\}$$

is called the **fundamental Weyl chamber**, an s-translate of which is called a **chamber**. The closure $\overline{C}$ is a convex polyhedral cone spanned by the subspace

$$F_B = \{x \in V : \check{\alpha}_i(x) = 0, 1 \leq i \leq r\}$$

and the vectors ω_i such that $\check{\alpha}_i(\omega_j) = \delta_{ij}$ (the Kronecker delta). The latter vectors are defined up to an element of F_B and are called the **fundamental weights**. The union of the closures of chambers, $I = \bigcup_{s \in W} s(\overline{C})$ is called the **Tits cone** of B. It is a convex cone in V. The Weyl group W acts totally discontinuously in the interior I^0 of I and $\overline{C}$ is a fundamental domain for this action. For any subset B' of B, let

$$F_{B'} = \{x \in V : \check{\alpha}_i(x) > 0, \alpha_i \in B \setminus B', \check{\alpha}_i(x) = 0, \alpha \in B'\}$$

be one of the faces of $\overline{C}$. Then the isotropy subgroup of a point $x \in \overline{C}$ is the group $W_{B'}$ generated by the s_i's belonging to the subset B' such that $x \in F_{B'}$ but $x \notin F_{B''}$ for any B'' containing B'. Then $W_{B'}$ is the Weyl group of the root base B' (for which the map $\vee$ is the restriction of the map $\vee$ defined for B). It follows from the above that the Weyl group W acts simply transitively on the set of chambers.

Main example: Let $f : X \to \mathbb{P}^k$ be a birational morphism of a nonsingular projective algebraic variety X onto the k-dimensional projective space which is a composition

$$X = X_0 \to X_1 \to \cdots \to X_r = \mathbb{P}^k$$

of monoidal transformations $f_i : X_{i-1} \to X_i$ centered at a point p_i. Let e_i' be the class of the exceptional space E_i of f_i in the 1-codimensional component $A^1(X_{i-1})$ of the Chow ring X_{i-1} (=the Picard group of $X_{i-1} = H^2(X_{i-1}, \mathbb{Z})$). Let h_i' be the class of a line in E_i in the 1-dimensional component $A_1(X_i)$ of the Chow ring of $X_{i-1}(= H_2(X_{i-1}, \mathbb{Z}))$. Let

$$e_i = (f_i \circ \cdots \circ f_1)^*(e_i') \in A^1(X),$$
$$h_i = (f_i \circ \cdots \circ f_1)^*(h_i') \in A_1(X),$$

for $1 \leq i \leq r$. And $e_0 = f^*(H)$ and $h_0 = f^*(h)$ where H (resp. h) is the class of a hyperplane (resp. a line) in $\mathbb{P}^k$. it follows from the standard intersection formulas that $\alpha_i = e_i - e_{i+1}$ and $\alpha_r = e_0 - e_1 - \cdots - e_{k+1}$ is a root basis in the space $V = A^1(X)_{\mathbb{R}} = A^1(X) \otimes_{\mathbb{Z}} \mathbb{R}$ with respect to the set

$$\check{B} = \{\check{\alpha}_1, \ldots, \check{\alpha}_r\} \subseteq A_1(X)_{\mathbb{R}} = V^*$$

where $\check{\alpha}_i = h_i - h_{i+1}$ and $\check{\alpha}_r = (k-1)h_0 - h_1 - \cdots - h_{k+1}$. It is easy to check that its Dynkin diagram is the graph $T_{k+1,r-k-1,2}$.

The simple reflection s_i acts on the basis $e_0, \ldots, e_r$ of $A^1(X)$ by leaving e_j $(j \neq i, i+1)$ fixed and permuting the e_i and e_{i+1}. The simple reflection s_r transforms e_0 to $ke_0 - e_1 - \cdots - e_{k+1}$ and e_i $(1 \leq i \leq k+1)$ to $(k-1)e_0 - e_1 - \cdots - e_{k+1} + e_i$, leaving all other e_i fixed. The closure $\overline{C}$ of the fundamental Weyl chamber is spanned by the space $F_B = \mathbb{R}K$, where $K = -(k+1)e_0 + (k-1)(e_1 + \cdots + e_r)$ is the canonical class of X, and the fundamental weights

$$\omega_1 = \frac{1}{k-1}e_0 - e_1,$$
$$\omega_2 = \frac{2}{k-1}e_0 - e_1 - e_2,$$
$$\cdots$$
$$\omega_k = \frac{k}{k-1}e_0 - e_1 - \cdots - e_k,$$
$$\omega_{k+1} = \frac{k+1}{k-1}e_0 - e_1 - \cdots - e_{k+1} = e_{k+2} + \cdots + e_r (\mathrm{mod} F_B),$$
$$\omega_{k+2} = e_{k+3} + \cdots + e_r,$$
$$\cdots$$
$$\omega_{r-1} = e_r,$$
$$\omega_r = \frac{1}{k-1}e_0.$$

One has to make a slight modification of these formulas if $k \geq r$. We can interpret geometrically some of these weights in the following way. Given a hypersurface of degree m in $\mathbb{P}^k$

with multiple points at $p_1, \dots, p_r$, of order $m_1, \dots, m_r$ resp. then its proper inverse transform in X belongs to the class $me_0 - m_1e_1 - \cdots - m_re_r$. Thus the weight ω_{k-1} represents the linear system of hyperplanes through the points $p_1, \dots, p_{k-1}$. The weight ω_{r-1} represents the exceptional divisor $E_r = f^{-1}(p_r)$. The $(k-1)$-multiple of ω_r represents the linear system of hyperplanes in $\mathbb{P}^k$.

Example 1: Four General Points. Let's blow up four points in $\mathbb{P}^3$ to create the space X_4^3. In this case $k = 3$ and $r = 4$. Then

$$A(X_4^3) = \mathbb{Z}[h, \pi_1, \pi_2, \pi_3, \pi_4]/(h^4, (h-\pi_i)^3, h\pi_i, \pi_i\pi_j)_{1\le i<j\le 4}.$$

Stratified by (co)dimension 1, we have

$$A^1(X_4^3) = \langle h, \pi_1, \pi_2, \pi_3, \pi_4\rangle$$

and

$$A_1(X_4^3) = A^2(X_4^3) = \langle h^2, -\pi_1^2, -\pi_2^2, -\pi_3^2, -\pi_4^2\rangle.$$

Then $\alpha_1 = \pi_1 - \pi_2$, $\alpha_2 = \pi_2 - \pi_3$, $\alpha_3 = \pi_3 - \pi_4$ and $\alpha_4 = h - \pi_1 - \pi_2 - \pi_3 - \pi_4$ is a root basis for $A^1(X_4^3)_{\mathbb{R}}$, and $\check{\alpha}_1 = -\pi_1^2 + \pi_2^2$, $\check{\alpha}_2 = -\pi_2^2 + \pi_3^2$, $\check{\alpha}_3 = -\pi_3^2 + \pi_4^2$, $\check{\alpha}_4 = 2h^2 + \pi_1^2 + \pi_2^2 + \pi_3^2 + \pi_4^2$.

Let us check $\check{\alpha}_i(\alpha_i) = -2$ for all i, where we have simple intersection product. Note that for $i < 4$, we have

$$(-\pi_i^2 + \pi_{i+1}^2)(\pi_i - \pi_{i+1}) = -\pi_i^3 - \pi_{i+1}^3 = -2h^3,$$

and for $i = 4$,

$$(2h^2 + \pi_1^2 + \pi_2^2 + \pi_3^4 + \pi_4^2)(h - \pi_1 - \pi_2 - \pi_3 - \pi_4) = 2h^3 - 4h^3 = -2h^3.$$

Since $\check{\alpha}_1(\alpha_2) = 1h^3 = \check{\alpha}_2(\alpha_3)$ and all others are 0, I guess our Dynkin diagram looks like

$$\alpha_1 \text{ —— } \alpha_2 \text{ —— } \alpha_3 \qquad \alpha_4$$

I think that makes sense for the diagram $T_{k+1,r-k-1,2} = T_{4,0,2}$, as purported: the 0 sort of "erases" the nucleus vertex.

We can also generate the reflections

$$s_i : A^1(X_4^3) \to A^1(X_4^3),\ x \mapsto x + \check{\alpha}_i(x)\alpha_i,$$

all of which fix the canonical class $K = -4h + 2\pi_1 + 2\pi_2 + 2\pi_3 + 2\pi_4$.

We have confirmed computationally that

- $\check{\alpha}_i(K) = 0$ for all i,
- $s_i(K) = K$ for all i,
- $s_i(\alpha_i) = -\alpha_i$ for all i,

- $s_i(\alpha_j) = \alpha_j$ for all $j \neq i$,
- $\alpha_i * K^2 = 0$ for all i.

We can also see that these are indeed reflections, as

$$\begin{aligned} s_i^2(x) = [x + \check{\alpha}_i(x)\alpha_i] + \check{\alpha}_i(x + \check{\alpha}_i(x)\alpha_i)\alpha_i &= x + \check{\alpha}_i(x)\alpha_i + (\check{\alpha}_i(x) + \check{\alpha}_i(x)\check{\alpha}_i(\alpha_i))\alpha_i \\ &= x + \check{\alpha}_i(x)\alpha_i + \check{\alpha}_i(x)\alpha_i - 2\check{\alpha}_i(x)\alpha_i = x, \end{aligned}$$

as well as verified computationally. We can build an alternate basis for $A^1(X_4^3)$ as

$$A^1(X_4^3) = \langle \alpha_1, \alpha_2, \alpha_3, \alpha_4, K \rangle = \check{K}^\perp \oplus \langle K \rangle$$

(note that $\langle \alpha_1, \alpha_2, \alpha_3, \alpha_4 \rangle$ is not really "$K^\perp$" because $\alpha_i * K \neq 0$, rather $\check{\alpha}_i(K) = 0$) with the change of basis matrix

$$\begin{pmatrix} 0 & 0 & 0 & 1 & -4 \\ 1 & 0 & 0 & -1 & 2 \\ -1 & 1 & 0 & -1 & 2 \\ 0 & -1 & 1 & -1 & 2 \\ 0 & 0 & -1 & -1 & 2 \end{pmatrix}^{-1} = \frac{1}{8} \begin{pmatrix} 0 & 6 & -2 & -2 & -2 \\ 0 & 4 & 4 & -4 & -4 \\ 0 & 2 & 2 & 2 & -6 \\ -8 & -4 & -4 & -4 & -4 \\ -4 & -1 & -1 & -1 & -1 \end{pmatrix}.$$

Note we seem to affirm Dolgachev's assertion that for $i < 4$, s_i fixes h and π_j for $j \notin i, i+1$, while s_i transposes π_i and π_{i+1}, but we **do not see** that $s_4(h) = 3h - \pi_1 - \pi_2 - \pi_3 - \pi_4$; instead,

$$s_4(h) = 3h - 2\pi_1 - 2\pi - 2\pi_3 - 2\pi_4.$$

Similarly, we **do not see** that $s_4(\pi_i) = 2h - \pi_1 - \pi_2 - \pi_3 - \pi_4 + \pi_i$; instead,

$$s_4(\pi_i) = h - \pi_1 - \pi_2 - \pi_3 - \pi_4 + \pi_i.$$

I'm not sure what the mistake is. I think the mistake is Dolgachev's; his matrix for s_4 would be

$$s_4 = \begin{pmatrix} 3 & 2 & 2 & 2 & 2 \\ -1 & 0 & -1 & -1 & -1 \\ -1 & -1 & 0 & -1 & -1 \\ -1 & -1 & -1 & 0 & -1 \\ -1 & -1 & -1 & -1 & 0 \end{pmatrix},$$

which sends K to $-K$; mine on the other hand is represented by the matrix

$$s_4 = \begin{pmatrix} 3 & 1 & 1 & 1 & 1 \\ -2 & 0 & -1 & -1 & -1 \\ -2 & -1 & 0 & -1 & -1 \\ -2 & -1 & -1 & 0 & -1 \\ -2 & -1 & -1 & -1 & 0 \end{pmatrix},$$

which fixes K.

Finally, we have the four fundamental weights:

- $\omega_1 = \frac{1}{2}h - \pi_1$,
- $\omega_2 = h - \pi_1 - \pi_2$,
- $\omega_3 = \frac{3}{2}\pi_1 - \pi_2 - \pi_3$,
- $\omega_4 = 2h - \pi_1 - \pi_2 - \pi_3 - \pi_4 \equiv \frac{1}{2}h \bmod F_B$.

I guess that's saying $s_4(h) = 3h - 2\pi_1 - 2\pi_2 - 2\pi_3 - 2\pi_4 \equiv 0 \bmod F_B$. Recall

$$F_B = \{x \in A^1(X_4^3)_{\mathbb{R}} : \check{\alpha}_i(x) = 0, 1 \le i \le r\}.$$

In this case that is we have the following kernels.

- $\ker \check{\alpha}_1 = \langle h, \pi_1 + \pi_2, \pi_3, \pi_4 \rangle$,
- $\ker \check{\alpha}_2 = \langle h, \pi_1, \pi_2 + \pi_3, \pi_4 \rangle$,
- $\ker \check{\alpha}_3 = \langle h, \pi_1, \pi_2, \pi_3 + \pi_4 \rangle$,
- $\ker \check{\alpha}_4 = \langle h - 2\pi_1, \pi_1 - \pi_2, \pi_1 - \pi_3, \pi_1 - \pi_4 \rangle$
- $F_B = \bigcap_{i=1}^{4} \ker \check{\alpha}_i = \langle 2h - \pi_1 - \pi_2 - \pi_3 - \pi_4 \rangle$.

So wait, $s_4(h) = 2(2h - \pi_1 - \pi_2 - \pi_3 - \pi_4) - h$, so actually $s_4(h) \equiv -h \bmod F_B$. Another mistake?

Example 2: Four General Lines. For our edification, let us consider the case of four general lines. We have

$$A(Y) = \mathbb{Z}[h, \lambda_1, \ldots, \lambda_4]/(h^4, (h - \lambda_i)^2, h\lambda_i(h - \lambda_i) - h^3, \lambda_i\lambda_j)_{1 \le i,j \le 4, i \ne j}.$$

Then $K = -4h + \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4$, and

$$A^1(Y) = \langle h, \lambda_1, \ldots, \lambda_4 \rangle$$

and

$$A_1(Y) = \langle h^2, h\lambda_1, \ldots, h\lambda_4 \rangle.$$

That is, our 1-dimensional curves can be generated by a general line and our "vertical" curves. The "horizontal" curves in each $\mathbb{P}^1 \times \mathbb{P}^1$ are linear combinations thereof: specifically,

$$(h - \lambda_i)\lambda_i = h^2 - h\lambda_i.$$

Then our roots are $\alpha_1 = \lambda_1 - \lambda_2$, $\alpha_2 = \lambda_2 - \lambda_3$, $\alpha_3 = \lambda_3 - \lambda_4$, and $\alpha_4 = 2h - \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4$ (note they are all zero-divisors with K^2) and their duals are $\check{\alpha}_1 = h\lambda_1 - h\lambda_2$, $\check{\alpha}_2 = h\lambda_2 - h\lambda_3$, $\check{\alpha}_3 = h\lambda_3 - h\lambda_4$, and $\check{\alpha}_4 = h^2 - h\lambda_1 - h\lambda_2 - h\lambda_3 - h\lambda_4$. Then the incidence relations are exactly the same as in the four points. In particular, we have the Dynkin diagram again $T_{4,0,2}$:

$$\alpha_1 \text{ —— } \alpha_2 \text{ —— } \alpha_3 \qquad \alpha_4$$

Again, all the $\check{\alpha}$'s are orthogonal to K. The simple reflections are also quite similar to those before: s_i transposes α_i and α_{i+1} for $1 \leq i < 4$, and

$$s_4 = \begin{pmatrix} 3 & 2 & 2 & 2 & 2 \\ -1 & 0 & -1 & -1 & -1 \\ -1 & -1 & 0 & -1 & -1 \\ -1 & -1 & -1 & 0 & -1 \\ -1 & -1 & -1 & -1 & 0 \end{pmatrix}.$$

We can also find the action of each reflection on the one-dimensional components. For example,

$$[s_4]_1 = \begin{pmatrix} 5 & 3 & 3 & 3 & 3 \\ -4 & -2 & -3 & -3 & -3 \\ -4 & -3 & -2 & -3 & -3 \\ -4 & -3 & -3 & -2 & -3 \\ -4 & -3 & -3 & -3 & -2 \end{pmatrix},$$

which also allows us to compute the images of the horizontal lines:

$$[s_4]_1((h - \lambda_i)\lambda_i) = 2h^2 - \left(\sum_{j=1}^{4} h\lambda_j\right) - h\lambda_i.$$

Note that unlike s_4, $[s_4]_1$ is not an involution: it does not even have finite order or a unital determinant. Maybe that means the tetrahedral roots should be

- $\pi_1 - \pi_2$
- $\pi_2 - \pi_3$
- $\pi_3 - \pi_4$
- $h - \pi_1 - \pi_2 - \pi_3 - \pi_4$
- $\lambda_{12} - \lambda_{13}$
- $\lambda_{13} - \lambda_{14}$
- $\lambda_{14} - \lambda_{23}$
- $\lambda_{23} - \lambda_{24}$
- $\lambda_{24} - \lambda_{34}$
- $2h - \lambda_{12} - \lambda_{13} - \lambda_{14} - \lambda_{23}$

No, that doesn't work.

3.1 Blowing up a Grassmannian

Let X be the blow up $Gr(2,4)$ at four points. Then

$$A(X) = A(Gr(2,4))[p_1,p_2,p_3,p_3]/(\sigma_a p_i, p_i p_j, (\sigma_1 - p_i)^4 - \sigma_{2,2})_{1\le i\le 4, i\neq j}.$$

Then since the canonical divisor of $Gr(2,4)$ is $-4\sigma_1$ by the adjunction formula (in general the canonical divisor of $Gr(k,n)$ is $-n\sigma_1$), and since $\dim Gr(2,4) = 4$, the canonical divisor of X is

$$K_X = -4\sigma_1 + 3p_1 + 3p_2 + 3p_3 + 3p_4.$$

Now we may make a root system $\alpha_1 = p_1 - p_2$, $\alpha_2 = p_2 - p_3$, $\alpha_3 = p_3 - p_4$, and $\alpha_4 = 2\sigma_1 - 2p_1 - 2p_2 - 2p_3 - 2p_4$. Then $\check{\alpha}_1 = p_1^3 - p_2^3$, $\check{\alpha}_2 = p_2^3 - p_3^3$ $\check{\alpha}_3 = p_3^3 - p_4^3$, and $\check{\alpha}_4 = 3\sigma_{2,1} - p_1^3 - p_2^3 - p_3^3 - p_4^3$. Notice that K^3 is not orthogonal to all of the α's, but K is indeed orthogonal to all of the $\check{\alpha}$'s. The simple reflection about α_4 is given by the matrix

$$\begin{pmatrix} 7 & 2 & 2 & 2 & 2 \\ -6 & -1 & -2 & -2 & -2 \\ -6 & -2 & -1 & -2 & -2 \\ -6 & -2 & -2 & -1 & -2 \\ -6 & -2 & -2 & -2 & -1 \end{pmatrix},$$

which one can check is indeed an involution that fixes K.

Note that four general points determine two Σ_1's. Four points in a particular position may determine one. One potential problem is that this transformation maps $\sigma_1 - p_1 - p_2 - p_3 - p_4$ (which is an effective divisor) to its own negative.

4 Postinghel's Slides

The Weyl group is generated by the Cremona transformations centered at the blowup points.

A Weyl divisor is a divisor in the orbit of one of the exceptional divisors E_i in the Weyl group. An effective cycle is a Weyl cycle if it is an irreducible component of an intersection of pairwise orthogonal (w.r.t. the Mukai-Dolgachev pairing) Weyl divisors.

The Mukai-Dolgachev pairing (for blowing up points in $\mathbb{P}^n$):

- $\langle H, H\rangle = n - 1$,
- $\langle H, E_i\rangle = 0$,
- $\langle E_i, E_j\rangle = -\delta_{ij}$.
- D is a (-1)-divisor if $\langle D, D\rangle = -1$ and $\frac{1}{n-1}\langle D, -K\rangle = 1$.